

Assignment No: 03

Descriptive Statistics

- Measures of Central Tendency and variability Perform the following operations on any open-source dataset (e.g., data.csv)
 - Provide summary statistics (mean, median, minimum, maximum, standard deviation) for a dataset (age, income etc.) with numeric variables grouped by one of the qualitative (categorical) variable. For example, if your categorical variable is age groups and quantitative variable is income, then provide summary statistics of income grouped by the age groups. Create a list that contains a numeric value for each response to the categorical variable.
 - Write a Python program to display some basic statistical details like percentile, mean, standard deviation etc. of the species of 'Iris-setosa', 'Iris-versicolor' and 'Iris-versicolor' of iris.csv dataset. Provide the codes with outputs and explain everything that you do in this step.

Import all required libraries

In [4]:

```
import io
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn import datasets

%matplotlib inline
```

In [7]:

```
df = pd.read_csv('data_students_performance.csv')
```

In [9]:

```
data_desc = pd.DataFrame(df.groupby('race/ethnicity').describe())
data_desc
```

Out[9]:

	math score									reading score					writing score	
	count	mean	std	min	25%	50%	75%	max	count	mean	...	75%	max	count	mean	
race/ethnicity																
group A	89.0	61.629213	14.523008	28.0	51.00	61.0	71.0	100.0	89.0	64.674157	...	74.00	100.0	89.0	62.67415	
group B	190.0	63.452632	15.468191	8.0	54.00	63.0	74.0	97.0	190.0	67.352632	...	79.75	97.0	190.0	65.60000	
group C	319.0	64.463950	14.852666	0.0	55.00	65.0	74.0	98.0	319.0	69.103448	...	78.50	100.0	319.0	67.82758	
group D	262.0	67.362595	13.769386	26.0	59.00	69.0	77.0	100.0	262.0	70.030534	...	79.00	100.0	262.0	70.14503	
group E	140.0	73.821429	15.534259	30.0	64.75	74.5	85.0	100.0	140.0	73.028571	...	84.00	100.0	140.0	71.40714	

5 rows x 24 columns

In [10]:

```
data_desc['math score']
```

Out[10]:

count	mean	std	min	25%	50%	75%	max
-------	------	-----	-----	-----	-----	-----	-----

race/ethnicity	count	mean	std	min	25%	50%	75%	max
group A	89.0	61.629213	14.523008	28.0	51.00	61.0	71.0	100.0
group B	190.0	63.452632	15.468191	8.0	54.00	63.0	74.0	97.0
group C	319.0	64.463950	14.852666	0.0	55.00	65.0	74.0	98.0
group D	262.0	67.362595	13.769386	26.0	59.00	69.0	77.0	100.0
group E	140.0	73.821429	15.534259	30.0	64.75	74.5	85.0	100.0

In [12]:

```
data_desc.transpose()
```

Out[12]:

	race/ethnicity	group A	group B	group C	group D	group E
math score	count	89.000000	190.000000	319.000000	262.000000	140.000000
	mean	61.629213	63.452632	64.463950	67.362595	73.821429
	std	14.523008	15.468191	14.852666	13.769386	15.534259
	min	28.000000	8.000000	0.000000	26.000000	30.000000
	25%	51.000000	54.000000	55.000000	59.000000	64.750000
	50%	61.000000	63.000000	65.000000	69.000000	74.500000
	75%	71.000000	74.000000	74.000000	77.000000	85.000000
	max	100.000000	97.000000	98.000000	100.000000	100.000000
reading score	count	89.000000	190.000000	319.000000	262.000000	140.000000
	mean	64.674157	67.352632	69.103448	70.030534	73.028571
	std	15.543762	15.177499	13.997033	13.895306	14.874024
	min	23.000000	24.000000	17.000000	31.000000	26.000000
	25%	53.000000	56.000000	60.000000	60.250000	63.000000
	50%	64.000000	67.000000	71.000000	71.000000	74.000000
	75%	74.000000	79.750000	78.500000	79.000000	84.000000
	max	100.000000	97.000000	100.000000	100.000000	100.000000
writing score	count	89.000000	190.000000	319.000000	262.000000	140.000000
	mean	62.674157	65.600000	67.827586	70.145038	71.407143
	std	15.468278	15.625173	14.983378	14.367707	15.113906
	min	19.000000	15.000000	10.000000	32.000000	22.000000
	25%	51.000000	55.250000	57.000000	61.000000	62.000000
	50%	62.000000	67.000000	68.000000	72.000000	72.000000
	75%	73.000000	78.000000	79.000000	80.000000	80.250000
	max	97.000000	96.000000	100.000000	100.000000	100.000000

Operations on iris dataset

In [14]:

```
from sklearn.datasets import load_iris
iris = load_iris()
```

In [20]:

```
iris_df = pd.DataFrame(iris.data)
```

```
iris_df['class'] = iris.target

iris_df.columns = ['sepal_len', 'sepal_wid', 'petal_len', 'petal_wid', 'class']

iris_df.head()
```

Out[20]:

	sepal_len	sepal_wid	petal_len	petal_wid	class
0	5.1	3.5	1.4	0.2	0
1	4.9	3.0	1.4	0.2	0
2	4.7	3.2	1.3	0.2	0
3	4.6	3.1	1.5	0.2	0
4	5.0	3.6	1.4	0.2	0

In [22]:

```
iris_df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 150 entries, 0 to 149
Data columns (total 5 columns):
#   Column      Non-Null Count  Dtype
---  -
0   sepal_len   150 non-null    float64
1   sepal_wid   150 non-null    float64
2   petal_len   150 non-null    float64
3   petal_wid   150 non-null    float64
4   class       150 non-null    int64
dtypes: float64(4), int64(1)
memory usage: 6.0 KB
```

In [25]:

```
iris_df['class'] = iris_df['class'].replace({0:'setosa',1:'versicolor',2:'virginica'})

iris_df.head()
```

Out[25]:

	sepal_len	sepal_wid	petal_len	petal_wid	class
0	5.1	3.5	1.4	0.2	setosa
1	4.9	3.0	1.4	0.2	setosa
2	4.7	3.2	1.3	0.2	setosa
3	4.6	3.1	1.5	0.2	setosa
4	5.0	3.6	1.4	0.2	setosa

In [27]:

```
data_desc = pd.DataFrame(iris_df.groupby('class').describe())
data_desc
```

Out[27]:

	sepal_len								sepal_wid			...	petal_len		petal_wid			
	count	mean	std	min	25%	50%	75%	max	count	mean	...	75%	max	count	mean	std	min	2
class																		
setosa	50.0	5.006	0.352490	4.3	4.800	5.0	5.2	5.8	50.0	3.428	...	1.575	1.9	50.0	0.246	0.105386	0.1	
versicolor	50.0	5.936	0.516171	4.9	5.600	5.9	6.3	7.0	50.0	2.770	...	4.600	5.1	50.0	1.326	0.197753	1.0	
virginica	50.0	6.588	0.635880	4.9	6.225	6.5	6.9	7.9	50.0	2.974	...	5.875	6.9	50.0	2.026	0.274650	1.4	

	sepal_len				sepal_wid				...	petal_len				petal_wid				
3 rows × 32 columns	count	mean	std	min	25%	50%	75%	max	count	mean	...	75%	max	count	mean	std	min	2
◀																▶		

In [29]:

```
data_desc.transpose()
```

Out[29]:

	class	setosa	versicolor	virginica
sepal_len	count	50.000000	50.000000	50.000000
	mean	5.006000	5.936000	6.588000
	std	0.352490	0.516171	0.635880
	min	4.300000	4.900000	4.900000
	25%	4.800000	5.600000	6.225000
	50%	5.000000	5.900000	6.500000
	75%	5.200000	6.300000	6.900000
	max	5.800000	7.000000	7.900000
sepal_wid	count	50.000000	50.000000	50.000000
	mean	3.428000	2.770000	2.974000
	std	0.379064	0.313798	0.322497
	min	2.300000	2.000000	2.200000
	25%	3.200000	2.525000	2.800000
	50%	3.400000	2.800000	3.000000
	75%	3.675000	3.000000	3.175000
	max	4.400000	3.400000	3.800000
petal_len	count	50.000000	50.000000	50.000000
	mean	1.462000	4.260000	5.552000
	std	0.173664	0.469911	0.551895
	min	1.000000	3.000000	4.500000
	25%	1.400000	4.000000	5.100000
	50%	1.500000	4.350000	5.550000
	75%	1.575000	4.600000	5.875000
	max	1.900000	5.100000	6.900000
petal_wid	count	50.000000	50.000000	50.000000
	mean	0.246000	1.326000	2.026000
	std	0.105386	0.197753	0.274650
	min	0.100000	1.000000	1.400000
	25%	0.200000	1.200000	1.800000
	50%	0.200000	1.300000	2.000000
	75%	0.300000	1.500000	2.300000
	max	0.600000	1.800000	2.500000

In [32]:

```
setosa_data = iris_df[iris_df['class'] == 'setosa']
versicolor_data = iris_df[iris_df['class'] == 'versicolor']
viginica_data = iris_df[iris_df['class'] == 'verginica']
```

In [33]:

```
setosa_data.describe()
```

Out[33]:

	selpal_len	sepal_wid	petal_len	petal_wid
count	50.00000	50.000000	50.000000	50.000000
mean	5.00600	3.428000	1.462000	0.246000
std	0.35249	0.379064	0.173664	0.105386
min	4.30000	2.300000	1.000000	0.100000
25%	4.80000	3.200000	1.400000	0.200000
50%	5.00000	3.400000	1.500000	0.200000
75%	5.20000	3.675000	1.575000	0.300000
max	5.80000	4.400000	1.900000	0.600000

In [37]:

```
versicolor_data.describe()
```

Out[37]:

	selpal_len	sepal_wid	petal_len	petal_wid
count	50.000000	50.000000	50.000000	50.000000
mean	5.936000	2.770000	4.260000	1.326000
std	0.516171	0.313798	0.469911	0.197753
min	4.900000	2.000000	3.000000	1.000000
25%	5.600000	2.525000	4.000000	1.200000
50%	5.900000	2.800000	4.350000	1.300000
75%	6.300000	3.000000	4.600000	1.500000
max	7.000000	3.400000	5.100000	1.800000

In [36]:

```
viginica_data.describe()
```

Out[36]:

	selpal_len	sepal_wid	petal_len	petal_wid
count	0.0	0.0	0.0	0.0
mean	NaN	NaN	NaN	NaN
std	NaN	NaN	NaN	NaN
min	NaN	NaN	NaN	NaN
25%	NaN	NaN	NaN	NaN
50%	NaN	NaN	NaN	NaN
75%	NaN	NaN	NaN	NaN
max	NaN	NaN	NaN	NaN

In []: